

BGIS 1ST SEMESTER

FINAL PROJECT



Blue Sky City:

Community livability Assessment for Calgary

GEOS 406 / 409 / 410 / 418 / 419

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EXECUTIVE SUMMARY

A community livability assessment of Calgary was conducted to determine the most livable communities within the city by analyzing housing affordability, transit accessibility, park coverage, and break-and-enter statistics. As Calgary continues to experience rapid population growth and changing economic conditions, understanding differences between neighbourhoods in terms of infrastructure, safety, and affordability has become increasingly important for residents and municipal planners. This project was completed using ArcGIS Pro and geospatial database methods to create a Calgary Livability Index that compares and ranks residential communities across the city. Data was obtained from multiple online sources including Pillar 9, the City of Calgary Open Data Portal, and Calgary Police Service to build a normalized spatial model of livability.

The results of the analysis indicate that livability in Calgary is not evenly distributed, but instead follows clear spatial patterns linked to established infrastructure and access to services. Communities such as Eau Claire and Downtown West End received some of the highest livability scores due to their strong transit connectivity, integrated green spaces, and overall accessibility. In contrast, lower scores were generally found in developing peripheral communities and high-value, low-density neighbourhoods where infrastructure and services are either limited or have not yet kept pace with residential growth. These findings provide a practical and data-driven resource for understanding the spatial distribution of urban well-being in Calgary and may be used as a guideline for future planning and community evaluation.

INTRODUCTION

BACKGROUND RESEARCH

Calgary's urban landscape has transformed significantly since the pandemic, as rapid population growth strains municipal infrastructure and housing markets. Sheikh and Biswas (2022) explain that livability is defined by how residents interact with their surroundings. As the city expands and zoning laws evolve, identifying these spatial disparities is essential for maintaining urban well-being.

GEOGRAPHIC QUESTION

"Where in Calgary, are the most livable communities?"

SOLUTION DESCRIPTION

The project develops a Calgary Livability Index within a geospatial framework. Using ArcGIS Pro, we created a normalized geodatabase integrating housing prices, park coverage, transit accessibility, and crime statistics. Because these indicators involve different units, we applied min-max scaling to standardize values and produce a consistent ranking for every neighborhood on a scale from 0 to 40 (esri, n.d.).

SOLUTION JUSTIFICATION

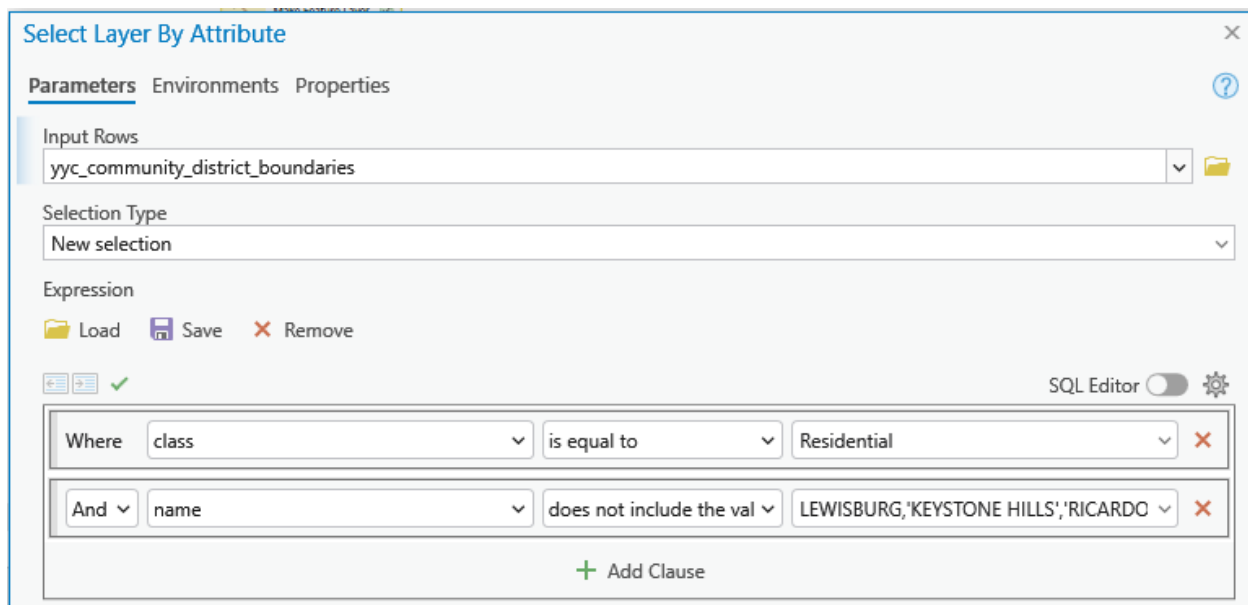
A spatial approach is necessary to evaluate geographic relationships such as walking distances and transit proximity. A normalized index ensures objective rankings by preventing any single factor from disproportionately influencing the final score. This data-driven resource assists planners and residents in making informed decisions regarding community equity and quality of life.

GEOS 406: GEOSPATIAL PROJECT FOUNDATIONS

PROJECT METHODOLOGY

(See Appendix E: Project Methodology Flow Chart)

The project methodology followed a logical progression from initial identification to final output. We defined four core livability pillars for Calgary: housing affordability, green space, transit accessibility, and safety. After processing the raw data into a central geodatabase, we refined the study area using attribute queries (Figure 1). This step filtered out industrial zones and institutional campuses, such as the University of Calgary, to focus strictly on residential neighborhoods. The final phase involved calculating and normalizing these variables into a comprehensive livability index for community level analysis.



Select Layer By Attribute

Parameters | Environments | Properties

Input Rows
yyc_community_district_boundaries

Selection Type
New selection

Expression

Load Save Remove

SQL Editor

Where class is equal to Residential

And name does not include the val LEWISBURG, KEYSTONE HILLS, RICARDO

+ Add Clause

Figure 1: Attribute query selection for refining the study area

PROJECT MANAGEMENT SUMMARY

Project management was anchored in an MS Project plan featuring a detailed Gantt chart (Appendix A) and Work Breakdown Structure. The workflow spanned from initial data acquisition in late February to final map production in early April. Key milestones, including the mid-term presentation on March 2, helped maintain momentum throughout the geodatabase setup and analysis phases. Resources were

strategically assigned to the team to ensure accountability and a balanced workload during the technical execution of the index.

RISK MANAGEMENT

Our risk management strategy (Appendix B) addressed the technical challenges inherent in multi-dataset projects. We identified potential issues such as community name mismatches, join errors, and topological inconsistencies. To mitigate these risks, the team enforced a shared folder structure and standardized coordinate systems across all layers. Frequent QA/QC audits allowed us to catch errors early in the process. Additionally, we addressed the challenge of time management by optimizing our file packaging workflow to occur only after major project updates.

GEOS 409: DATA CAPTURE

DATA SOURCES

Source	Data Provided	Purpose in Project
City of Calgary Open Data Portal (City of Calgary, n.d.)	Community boundaries, parks, land use, roads, transit networks	Provided core spatial layers for mapping and spatial analysis
Pillar 9 (9, n.d.)	2025 housing prices	Used to analyze housing affordability across Calgary communities
Calgary Police Service (Calgary Police Service, n.d.)	Public safety / crime data	Used to evaluate safety and livability metrics

Table 1: Project Data Sources

TOPOLOGY

To maintain spatial integrity, strict topology rules were applied to the Communities_Final layer. We verified that the polygons contained no gaps and no overlaps, ensuring a continuous representation of Calgary's neighborhoods. Additionally, a "Must Be Covered By" rule linked the communities to the City_Boundary layer to guarantee all analysis remained within the municipal limits. These quality control steps were vital for accurate area-based calculations.

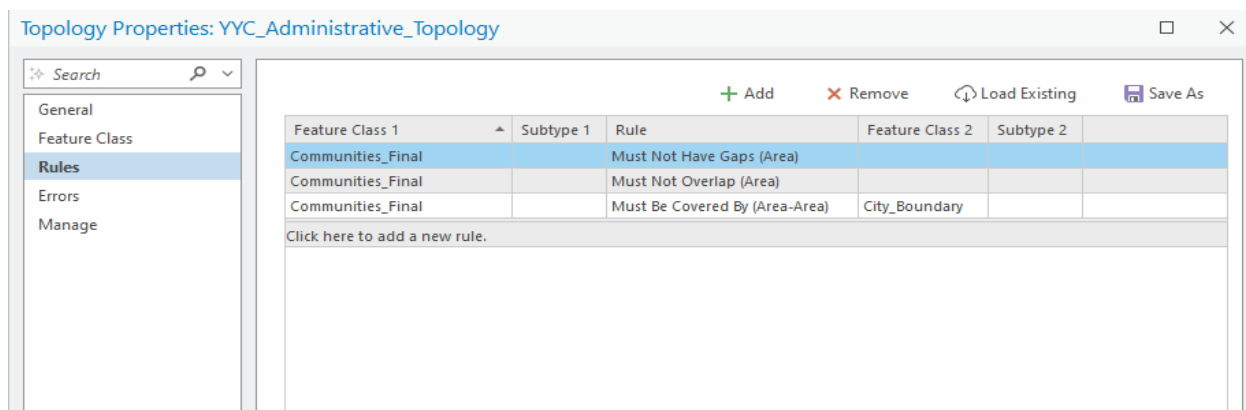


Figure 2: Topology Rules Applied to Communities_Final Layer

DATA CREATION

Two specific data creation techniques significantly enhanced the analysis. First, the community layer was enriched with 2025 population and median income statistics. This enabled the calculation of standardized metrics, such as break and enter rates per 1,000 residents, which provides a more accurate reflection of safety by mitigating population size bias (Department of Justice Canada, 2022).

Second, we developed a transit accessibility model utilizing a pedestrian network. A 10-minute walking service area was generated around transit and LRT stops, reflecting realistic commuter behavior (Statistics Canada, 2023; Via Transportation, n.d.). Assuming a 5 km/h walking speed and treating major roads as barriers, these service areas were intersected with community boundaries to determine the precise percentage of each neighborhood within walking distance of public transit. This approach mirrors established municipal planning standards for evaluating urban mobility (City of Calgary, n.d.).

GEOS 410: DATA MANIPULATION AND TRANSFORMATION

FOLDER STRUCTURE AND PROJECT DIRECTORY

The project follows a well-organized folder structure within Group1_FinalProject to ensure efficient data management and minimize path-related errors. All original datasets are stored in the raw_data folder, while processed data and analysis outputs are maintained in bluesky.gdb. Model Builder outputs are stored separately in BlueSkyModel.gdb, and geoprocessing tools are organized within the bluesky_toolbox. Only finalized and cleaned layers are retained in the geodatabases, creating a clear and structured workflow. This organization aligns with the project README and supports full reproducibility, allowing the analysis to be traced from raw data through to final outputs.

FME/MODELBUILDER AUTOMATION

Workflow automation was completed using Model Builder in ArcGIS Pro to create a repeatable process from raw data to base layer. The workflow began by defining and projecting community boundary data to NAD 1983 3TM 114 for consistency.

A Select Layer by Attribute filtered residential communities and removed those with a population under 100. The Enrich tool added key variables such as population and median household income.

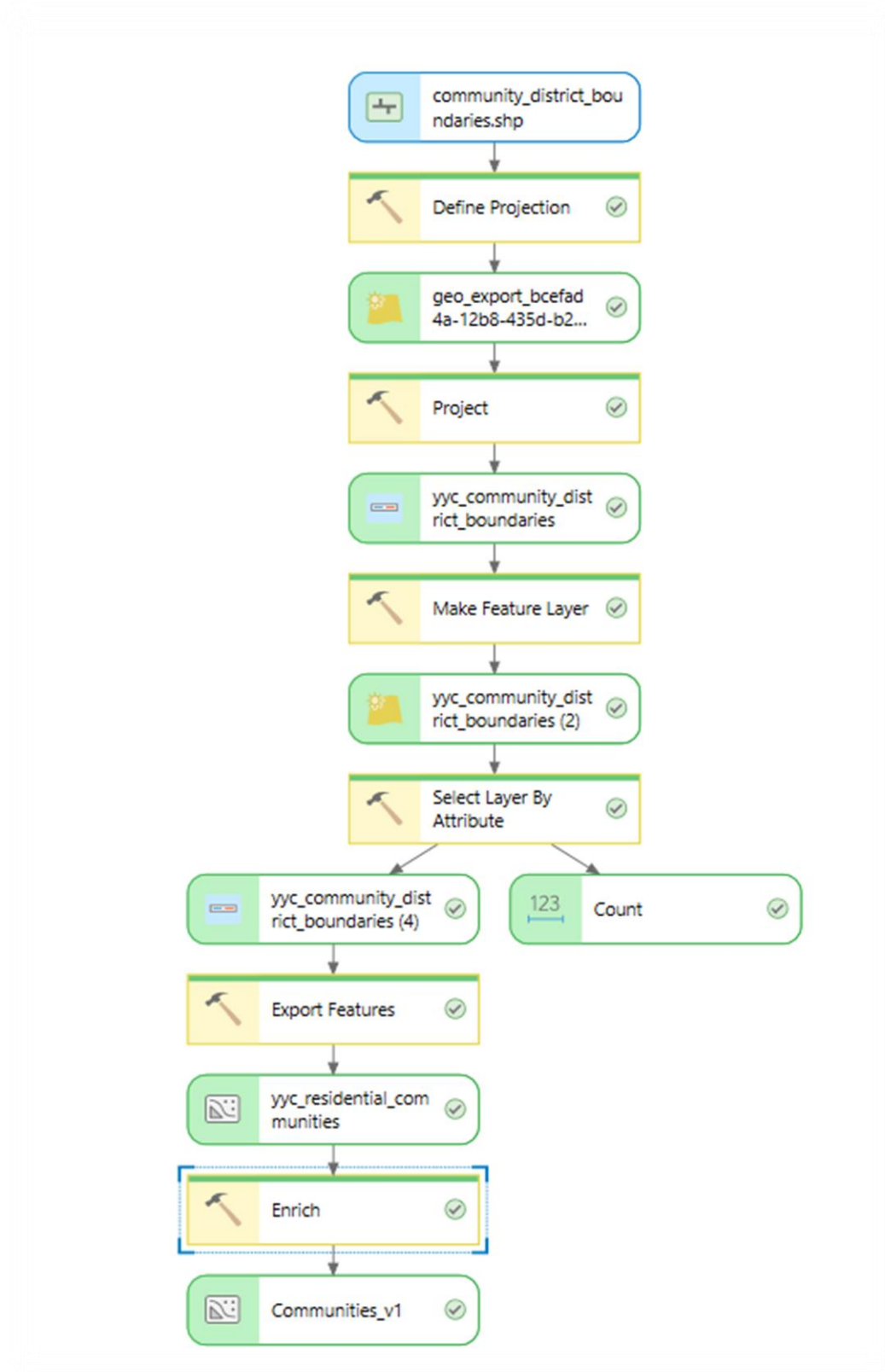


Figure 3: Model Builder Demonstrate Study Area Creation

CHECKLIST

QA/QC checks were completed at different stages using the project checklist (Appendix C: GEOS 410 Checklist: The completed QA/QC Excel checklist). This included ensuring all datasets were in the correct coordinate system, verifying joins (such as housing and crime data), and checking for missing or incorrect values. Polygon layers were also reviewed to ensure no overlaps or gaps after applying topology rules. Final maps were checked for clear symbology, labels, and key layout elements such as title, legend, and north arrow.

GEODATABASE DEVELOPMENT

A file geodatabase (bluesky.gdb) was developed to store all processed project data. Feature datasets such as YYC_Housing, YYC_Parks, YYC_Transportation, and YYC_Index were used to organize layers including Parks_in_Comm, Transit_ServiceArea, and YYC_Comm_CompositeIndex.

GEOS 418: DATA MODELLING

GEODATABASE & SPATIAL DATABASE DESIGN

The project database utilizes an ESRI File Geodatabase to organize spatial and tabular data into a single container. As shown in the ERD (Appendix D: GEOS 418 Data Dictionary), the schema consists of seven interrelated tables with Communities_Final as the central hub. This primary table stores essential attributes like crime rates, transit accessibility, and final livability scores. Records are identified by an OBJECTID primary key, using Comm_name as the common foreign key to link thematic datasets for housing and infrastructure. This preserves the original relational structure and coordinate reference system, supporting both desktop workflows and SQL-based spatial queries within a unified environment.

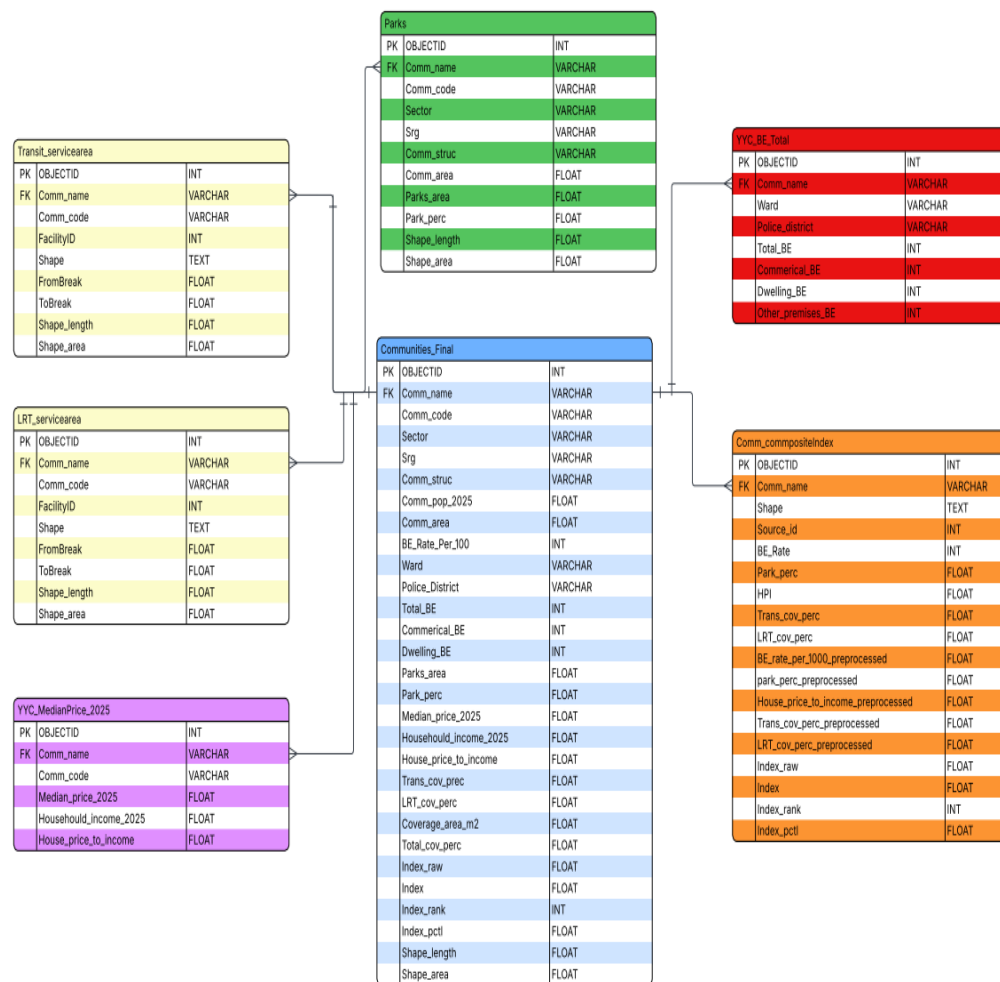


Figure 4: Entity Relationship Diagram (Appendix F)

GEOS 419: DATA ANALYSIS AND OUTPUT

SPATIAL ANALYSIS

The spatial analysis was carried out at the community level using the filtered Communities layer as the main unit of analysis. Each of the four livability pillars was processed separately using tools such as **Intersect, Spatial Join, and network analysis**, then linked back to create the final community layer. After preparation and normalization, these variables were mapped to examine the spatial distribution of livability conditions across Calgary.

Crime (B&E)

Crime was measured using break and enter rates per 1,000 residents. The map (**Appendix G**) results showed that crime was not distributed evenly across Calgary, with higher rates concentrated mainly in the central part of the city and lower rates more common in the north and southeast. This pattern also generally follows the LRT corridor. Previous research suggests this may occur because transit-accessible areas create greater opportunity and movement for offenders (Azad, 2011). The total B&E occurrences also support this pattern. The LRT lines intersect neighborhoods in the Northwest, Center, Northeast and south which are the sectors where the most total break and enter crimes occurred. Within the livability index, lower break and enter rates were treated as more favorable.

	OBJECTID *	sector	Total B&Es ▼
1	1	CENTRE	2794
2	6	SOUTH	1200
3	4	NORTHEAST	871
4	5	NORTHWEST	785
5	8	WEST	698
6	3	NORTH	645
7	7	SOUTHEAST	475
8	2	EAST	459

Figure 5: Total Break and Enters by Sector

Transportation Access

Transportation access (**Appendix H**) was measured using the percentage of each community located within a 10-minute walking service area of transit stops and LRT stations. Transit and LRT access were first measured separately, then combined into one transportation category with equal weighting. The results showed that many communities had strong transit coverage, with 64% (135 of 212) having more than 90% of their area within the walking threshold. The weakest coverage occurred mainly in peripheral developing communities such as Pine Creek, Rangeview, and Moraine, where transit infrastructure is still limited. Access to the high-capacity LRT stations was available to only 76 communities. Further to that, only 6 had over 80% coverage. In the livability model, higher transportation coverage was considered more favorable.

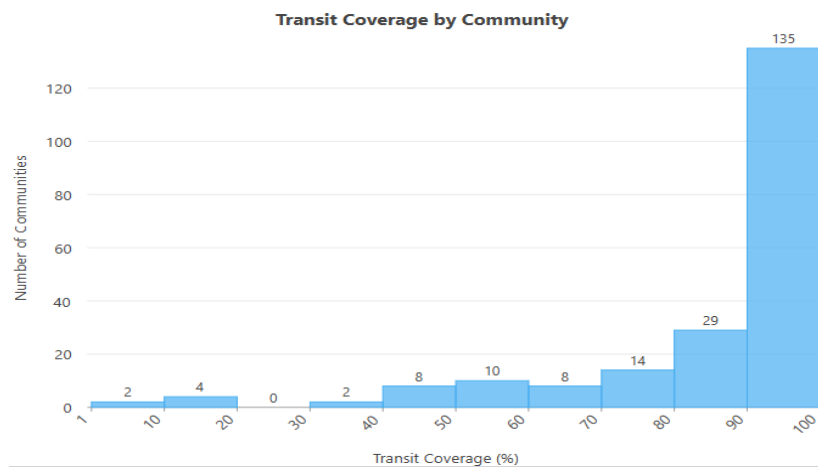


Figure 6: Percent coverage of communities by 10-minute transit service area

Housing Affordability

Housing affordability (**Appendix I**) was measured using the housing price to income ratio (HPI). Lower HPI values represent more affordable conditions, while higher values indicate a greater housing cost burden. The results showed a noticeable spatial pattern, with lower affordability concentrated in parts of the central city and in several outward-extending pockets, while many western and south/southeast peripheral communities showed more favorable affordability conditions. Overall, the map indicates that housing affordability is spatially clustered rather than randomly distributed across the city. In the final index, lower HPI values were reverse scored so that more affordable communities received higher livability scores.

Park Space

The park space map (**Appendix J**) shows clear spatial variation in the percentage of communities dedicated to parks across Calgary. Higher Park coverage appears in several western, central, and southern communities, while lower park percentages are more common in many eastern and outer-edge communities. This suggests that access to neighborhood-level green space is not evenly distributed across the city. In the livability index, communities with higher park coverage were considered more favorable.

The Livability Index

The final livability index map (**Appendix K**) shows that livability is not distributed evenly across Calgary but instead follows a clear spatial pattern. Higher index values are concentrated mainly in the central part of the city, with additional clusters extending into the northwest, west, and portions of the south. In contrast, lower livability scores are more common in peripheral communities, especially along parts of the eastern, northeastern, and far southern edges of Calgary. This suggests that livability is spatially clustered rather than randomly distributed. A noticeable concentration of higher scoring communities occurs in and around the Centre sector, where several neighbourhoods fall within the upper index classes. This pattern likely reflects the combined influence of stronger transit access, more established infrastructure, and closer proximity to services and amenities. Additional pockets of relatively high livability also appear in the Northwest and West sectors, although these are more dispersed than the central cluster. Lower scoring communities appear more frequently near the city margins, where lighter index classes are more common. Overall, the map indicates that communities with stronger livability conditions tend to occur in more established and better-connected areas, while lower scoring communities are more often associated with outer areas where service access remains limited.

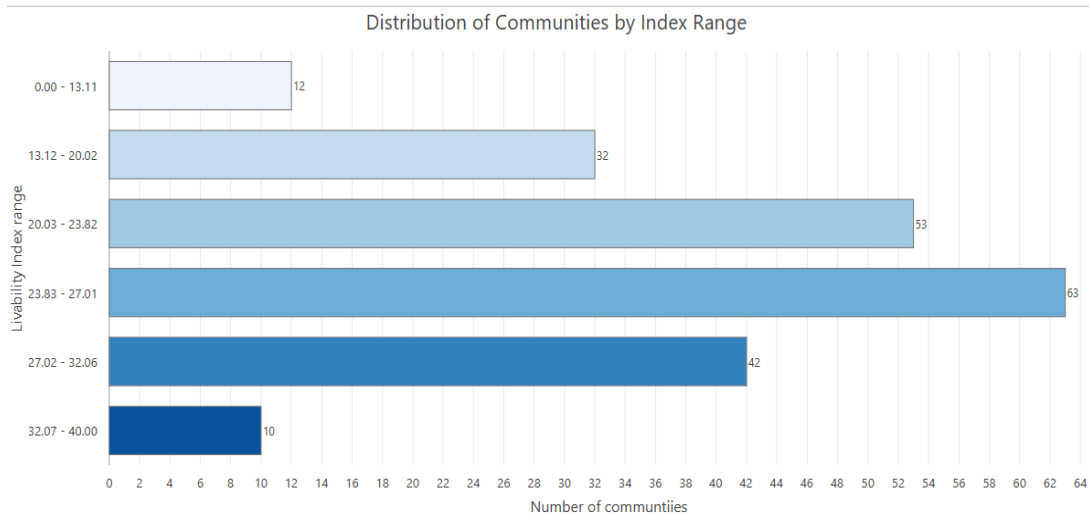


Figure 7: Graph of community distribution by Index range

DESCRIPTIVE STATISTICS

The statistical distribution of the final Livability Index provides a general overview of how livability is distributed across Calgary communities. On a scale from 0 to 40, the mean index value for the city was 23.83, while the median was slightly higher at 24.31. The small difference between these two values suggests that the distribution is relatively balanced overall, although it displays a slight negative skew.

This pattern indicates that many Calgary communities are grouped within the middle to upper portion of the index, particularly between scores of 20 and 30. The lower end of the distribution forms a smaller tail, representing a limited number of communities that perform less favorably in terms of accessibility, amenities, or overall livability conditions. With a standard deviation of 5.77, the index shows enough variation to distinguish stronger and weaker performing communities without becoming excessively dispersed. The lack of major outliers also suggests that the rankings provide a reasonably stable basis for comparing community well-being and identifying areas where future improvement may be needed.

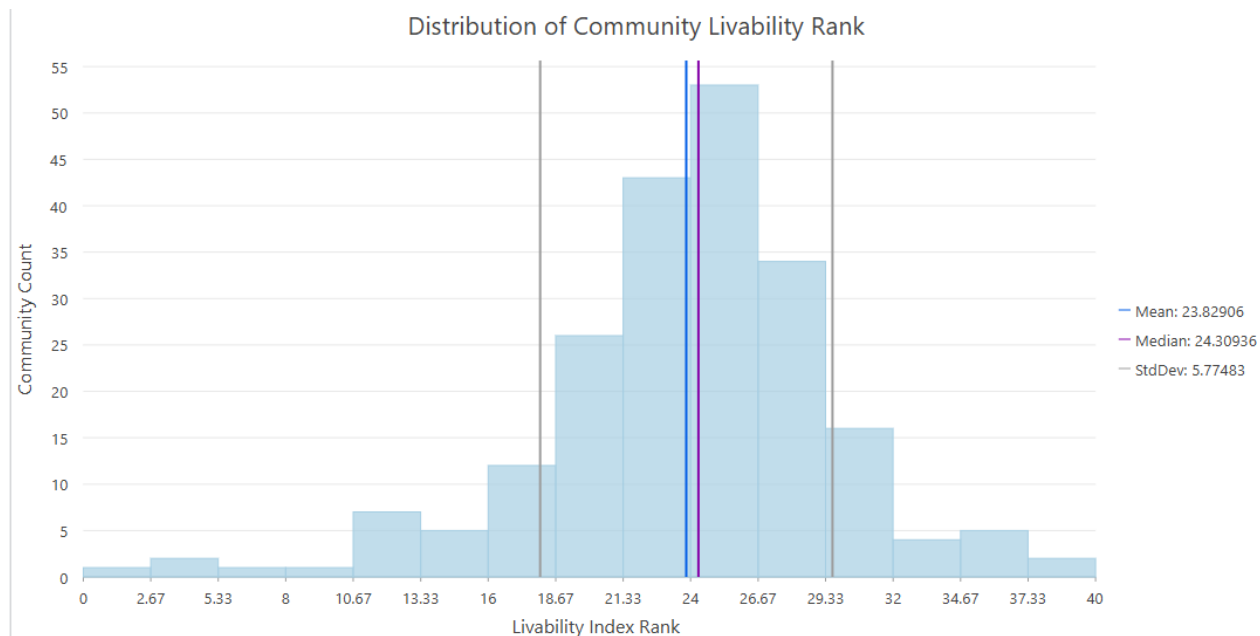


Figure 8: Distribution of Community Livability Rank.

ADVANCED SPATIAL ANALYSIS TECHNIQUES

Spatial Autocorrelation

To determine whether the index values we computed were statistically significant, we started with the Global Moran's I Spatial Autocorrelation tool. We applied Continuity edges corners for the community polygons as this spatial relationship defines touching communities as neighbors. The analysis returned a Moran's I value of 0.209074, a z-score of 4.819361, and a p-value of 0.000001. These results indicate statistically significant positive spatial autocorrelation, meaning that communities with similar livability scores tend to be located near one another. Because the p-value is extremely small and the z-score is well above the threshold for significance, the observed clustered pattern is highly unlikely to be the result of random chance. This confirms that the livability in Calgary as defined by this model is spatially structured rather than randomly distributed.

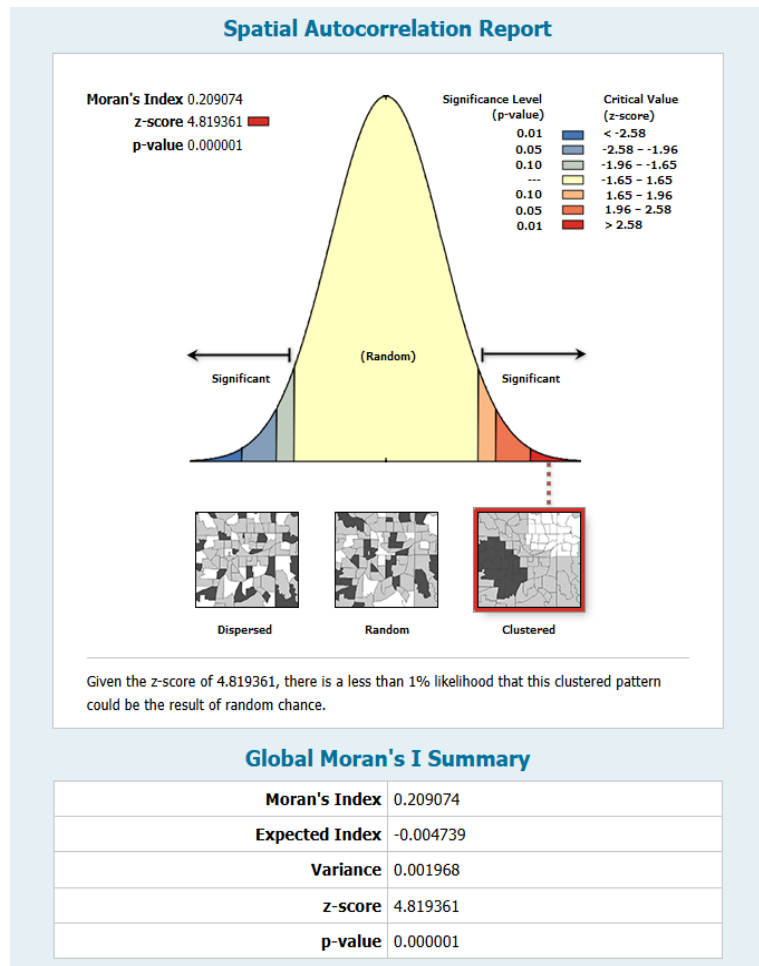


Figure 9. Global Moran's I result for Calgary Livability Index

Hot Spot Analysis

The Getis-Ord_Gi* tool was then used to identify the location of statistically significant clusters of high and low livability scores. The hot spot map (**Appendix L**) showed a concentration of high-value clusters in and around the central city, with additional smaller hot spot areas in parts of the west and northwest. Significant cold spots were identified in parts of the east and in smaller areas to the south. These results indicate that communities with stronger livability scores tend to cluster together in specific parts of Calgary, while lower-scoring communities also form localized clusters elsewhere in the city. The hot spot analysis therefore supports the conclusion that livability conditions are spatially grouped rather than evenly distributed.

Cluster and Outlier Analysis

Cluster and Outlier Analysis using Anselin Local Moran's I provided a more detailed local interpretation of the livability pattern. The map (**Appendix M**) identified several high-high clusters, representing communities with high livability scores surrounded by similarly high-scoring communities, as well as low-low clusters, where lower livability scores were concentrated among neighboring communities. A small number of spatial outliers were also present, indicating communities whose livability scores differed from those of the surrounding area. This is important because it shows that the Calgary livability pattern is not only clustered overall but also includes local exceptions where individual communities perform differently than nearby communities.

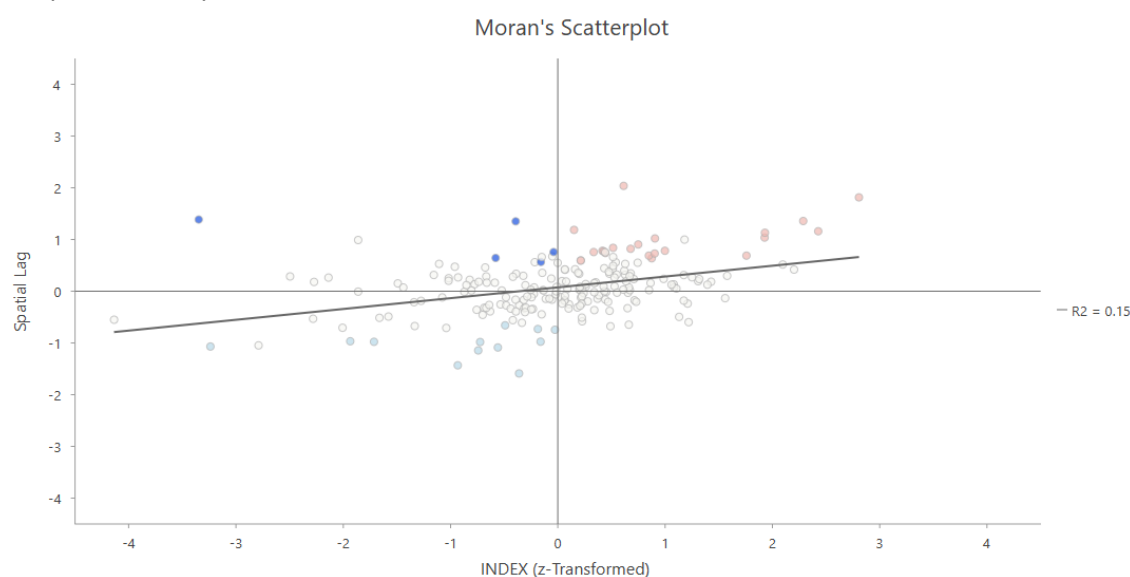


Figure 10: Moran's scatterplot showing a positive spatial autocorrelation

Interpretation and Results

The patterns identified in the maps explain why certain areas are more livable. The analysis shows a clear relationship between variables. High livability clusters are almost always found in areas with high transit accessibility and low residential crime. Communities that score low often suffer from a lack of usable green space or higher housing costs.

Following the objective approach suggested by Bochman (2016), the team let the data tell the story. The results confirm that safety and mobility are the strongest drivers of neighborhood satisfaction in Calgary. By moving beyond preconceived assumptions, the index provides a verifiable look at how infrastructure and security impact the urban experience.

CARTOGRAPHIC OUTPUT

Thematic maps are the primary tool for an effective visual answer to the "where" of this analysis. The layouts were designed for clarity to display the spatial distribution patterns that support the evidence-based conclusions of the report.

A choropleth map was used to display the final Livability Index across over 200 community polygons. Graduated colors are utilized for a quick comparison of neighborhood rankings across the entire city.

The data is classified using the Natural Breaks (Jenks) method and 6 classes. This classification was chosen because it identifies the inherent groupings and natural gaps within the scores, validating the classification with a GVF calculation above 0.9. This provides an accurate and objective reflection of the real-world differences between Calgary's diverse communities.

```
Feature Class: Communities_Final
Field: INDEX
Class Breaks (raw): ['13.11', '20.02', '23.82', '27.01', '32.06', '40']
Classification Method (optional): Natural Breaks (Jenks)
Validated and sorted class breaks: [13.11, 20.02, 23.82, 27.01, 32.06, 40.0]
Largest break is 40.0
Filtered Value Range: 0.00 - 40.00
GVF for custom classification (filtered values only): 0.9239
```

Figure 11: GVF calculation for final index.

	Name	Classification	Symbolization	Color schemes
1	Transportation Access		Unique Symbology	Light Blue and Dark Blue
2	HPI	Natural Breaks (Jenks)	Graduated Colors	Purple-Blue-Green (6 Classes)
3	Park Space	Natural Breaks (Jenks)	Graduated Colors	Yellow- Green (Continuous)
4	Crime(B&E)	Natural Breaks (Jenks)	Graduated Colors	Orange-Red (Continuous)
5	Livability Index	Natural Breaks (Jenks)	Graduated Colors	Blues (6 classes)

Table 2: Project Cartography

CONCLUSION

The Calgary Livability Index offers a clear, data-driven picture of quality of life across the city. By measuring safety, mobility, affordability, and green space, the data reveals that livability directly reflects infrastructure maturity rather than random chance.

Top ranking neighborhoods like the Beltline, Downtown West End, and Sunnyside thrive by providing a complete urban experience. Clustered near the city core, these communities benefit from historical infrastructure investments. Their high scores stem from a sophisticated balance of reliable transit, high walkability, established parks, and managed safety.

In contrast, peripheral developing communities like Pine Creek, Rangeview, and Moraine face a distinctly different reality. These areas often experience a service lag where rapid residential growth has outpaced essential urban infrastructure. Despite offering modern housing, they score lower due to limited public transit, elevated housing costs relative to income, and fewer mature parks. This spatial divide highlights that new development alone does not automatically guarantee high livability.

Ultimately, the index demonstrates that true livability is a cumulative achievement requiring a balance of all urban pillars. To bridge this geographic gap, city planners must prioritize equitable investments. Accelerating transit expansion in developing sectors, improving safety along commuter corridors, and ensuring fair access to green space will help reduce spatial inequality. This data-guided approach ensures that future investments are placed where they can most effectively improve the daily urban experience.

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Appendix B: GEOS 406 Documents: Risk Management Plan

Risk	Result/Impact on Project	Likelihood	Impact	Mitigation/Solution
Data loss during project work	Loss or corruption of spatial datasets, analysis results, or geodatabases could significantly delay the project and require reprocessing of data.	Moderate	High	Maintain multiple backups of raw and processed datasets. Store project files in shared cloud storage and create copies of datasets after each major processing step.
Team member leaving or being unavailable	If a key team member becomes unavailable, project knowledge, workflows, or data access could be lost which may delay project progress.	Low	High	Document all workflows, analysis steps, and login/access information in a shared document. Ensure all project data is stored in a shared folder so other members can continue the work.
Poor version control of project files	Multiple team members editing the same datasets or documents may create conflicting versions and confusion about which file is the latest.	High	High	Create a master project folder and implement check-in/check-out procedures. Assign one master ArcGIS project file and maintain clear file naming conventions with timestamps.

Inconsistent or incomplete datasets	Missing housing data, incomplete open datasets, or missing community attributes may result in some communities being excluded from analysis.	Moderate	High	Validate datasets before analysis and cross-check attributes. If data gaps exist, document limitations and adjust analysis to maintain consistency.
Temporal inconsistencies between datasets	Housing, crime, parks, and transit datasets may originate from different years which could produce misleading livability comparisons.	High	High	Use datasets from similar time periods when possible and document dataset years in metadata. Normalize values to reduce temporal bias.
Errors in data normalization or index calculations	Incorrect calculations when performing min-max scaling or livability scoring may distort the final ranking of communities.	Low	High	Test calculations on sample data first and validate outputs manually. Maintain clear documentation of the analysis workflow.
Incorrect spatial alignment or coordinate system issues	Datasets using different coordinate systems may not overlay correctly, causing spatial analysis errors.	Low	High	Standardize all datasets to a consistent coordinate system (e.g., NAD 83 UTM Zone 11N) before analysis.
Bias or assumptions in	The chosen variables (housing prices, green space, crime, transit	Low	Moderate	Support variable selection through literature review and clearly explain

selected variables	access) may not fully represent livability and could influence the interpretation of results.			assumptions in the methodology section.
Project packaging and repackaging delays (NEW)	Frequent updates to geodatabases and ArcGIS projects require repeated packaging, consuming time and risking inconsistencies.	Moderate	Moderate	Maintain a master packaged project, only repackage after major updates, and use version naming.
Simplification of indicators used in the livability model	Using simplified indicators such as B&E crime only, green space size instead of quality, or house prices as affordability proxies may oversimplify complex social factors.	Moderate	Moderate	Explain indicator limitations in the report and apply statistical techniques (such as median housing prices) to reduce bias from extreme values.

Appendix C: GEOS 410 Checklist: The completed QA/QC Excel checklist

GEOS410 Group 1– Data Integrity & Manipulation Geodatabase QA/QC Checklist						
BLUE SKY NAD 1983 3TM 114W						
Reviewers: Teng Zhang, Apram Singh & Ifechukwudelu Maduekwe			Date: 12/04/2026			
Section / Item	Status	Comments	Reviewed By	Approved By	Sign-off	Date
1. Geodatabase – General						
Geodatabase name is clear, concise and descriptive	✓ Approved	bluesky	TZ	AS	TZ, AS, IDM	04/12/2026
Naming convention followed (CamelCase or snake_case)	✓ Approved	snake_case	TZ	AS	TZ, AS, IDM	04/12/2026
GDB stored close to the root directory (≤ 2 folders deep)	✓ Approved	2 Folder Deep	TZ	AS	TZ, AS, IDM	04/12/2026
Only ONE final geodatabase submitted	✓ Approved	bluesky.gdb	TZ	AS	TZ, AS, IDM	04/12/2026
All intermediate / redundant data removed	✓ Approved	Only final layers kept in bluesky.gdb	TZ	AS	TZ, AS, IDM	04/12/2026
Metadata created and complete (Item Description format)	✓ Approved	Item Description completed for final datasets	TZ	AS	TZ, AS, IDM	04/12/2026
GDB is free from corruption / opens without errors	✓ Approved	Opened in ArcGIS Pro with no errors	TZ	AS	TZ, AS, IDM	04/12/2026
2. Feature Datasets						
Feature dataset names are clear, concise and descriptive	✓ Approved	ArcGIS Dataset	IDM	TZ	TZ, AS, IDM	04/12/2026
Naming convention followed (CamelCase or snake_case)	✓ Approved	snake_case	IDM	TZ	TZ, AS, IDM	04/12/2026
Spatial reference set to NAD 1983 3TM 114W	✓ Approved	NAD 1983 3TM 114W	IDM	TZ	TZ, AS, IDM	04/12/2026
All related feature classes grouped within correct dataset	✓ Approved	grouped by Admin, Crime, Housing, Parks, Transport	IDM	TZ	TZ, AS, IDM	04/12/2026
Metadata present for each feature dataset	✓ Approved	Metadata added for main feature datasets	IDM	TZ	TZ, AS, IDM	04/12/2026
No duplicate or redundant feature dataset	✓ Approved	No duplicate feature datasets in final GDB	IDM	TZ	TZ, AS, IDM	04/12/2026
3. Feature Classes – Structure & Naming						
Feature class names are clear and descriptive	✓ Approved	Names clearly describe layer content	TZ	IDM	TZ, AS, IDM	04/12/2026
Naming convention consistent (CamelCase or snake_case)	✓ Approved	snake_case	TZ	IDM	TZ, AS, IDM	04/12/2026
Geometry type is correct (Point / Polyline / Polygon / Raster)	✓ Approved	Point, line, polygon, and table types verified	TZ	IDM	TZ, AS, IDM	04/12/2026
Spatial reference set to NAD 1983 3TM 114W	✓ Approved	Verified in layer properties	TZ	IDM	TZ, AS, IDM	04/12/2026
Metadata present and complete for each feature class	✓ Approved	Metadata added for key final feature classes	TZ	IDM	TZ, AS, IDM	04/12/2026
No redundant or duplicate feature classes	✓ Approved	Only final feature classes retained	TZ	IDM	TZ, AS, IDM	04/12/2026
4. Attribute Tables						
Field names are descriptive and follow naming convention	✓ Approved	Field names are clear and consistent	AS	TZ	TZ, AS, IDM	04/12/2026
Correct data types assigned to all fields	✓ Approved	Text, numeric, and date fields verified	AS	TZ	TZ, AS, IDM	04/12/2026
No null / missing values where data should exist	✓ Approved	No critical nulls in final analysis fields	AS	TZ	TZ, AS, IDM	04/12/2026
No duplicate records in attribute table	✓ Approved	No duplicate records after joins	AS	TZ	TZ, AS, IDM	04/12/2026
Domain values are valid and within expected range	✓ Approved	Values checked against expected ranges	AS	TZ	TZ, AS, IDM	04/12/2026
Attribute table is clean and free of junk/legacy fields	✓ Approved	Temporary and legacy fields removed	AS	TZ	TZ, AS, IDM	04/12/2026
All joins verified – no unwanted duplicate records introduced	✓ Approved	Housing and crime joins checked for duplicates	AS	TZ	TZ, AS, IDM	04/12/2026
5. Topology & Data Quality						
Topology rules defined where applicable	✓ Approved	Topology rules created for communities and parks	IDM	TZ	TZ, AS, IDM	04/12/2026
Must Not Overlap rule checked and errors resolved	✓ Approved	Overlap errors reviewed and resolved	IDM	TZ	TZ, AS, IDM	04/12/2026
Must Not Have Gaps rule checked and errors resolved	✓ Approved	Gap checks completed for polygon layers	IDM	TZ	TZ, AS, IDM	04/12/2026
No invalid or null geometries	✓ Approved	Geometry checked and no null shapes found	IDM	TZ	TZ, AS, IDM	04/12/2026
Topology validated and all errors fixed or flagged	✓ Approved	Topology validated before final output	IDM	TZ	TZ, AS, IDM	04/12/2026
Features are spatially correct and within study area extent	✓ Approved	All features fall within Calgary study area	IDM	TZ	TZ, AS, IDM	04/12/2026
6. Projection & Coordinate System						
All datasets projected to NAD 1983 3TM 114W	✓ Approved	All final layers in NAD 1983 3TM 114W	TZ	AS	TZ, AS, IDM	04/12/2026
Projection verified in ArcGIS Pro layer properties	✓ Approved	Confirmed in ArcGIS Pro properties	TZ	AS	TZ, AS, IDM	04/12/2026
7. Redundancy & File Management						
No backup / temp layers left in final GDB	✓ Approved	Backup and scratch layers excluded from final GDB	TZ	IDM	TZ, AS, IDM	04/12/2026
All raster datasets properly named and stored	✓ Approved	Raster-related outputs named consistently	TZ	IDM	TZ, AS, IDM	04/12/2026
GDB compacted before final submission	✓ Approved	Geodatabase compacted before submission	TZ	IDM	TZ, AS, IDM	04/12/2026
TOTAL ITEMS APPROVED		37	0	← Auto-counted from Status column		

GEOS410 Group 1– Data Integrity & Manipulation Map Layout QA/QC Checklist						
BLUE SKY NAD 1983 3TM 114W						
Reviewers: Teng Zhang, Apram Singh & Ifechukwudelu Maduekwe			Date: 12/04/2026			
Section / Item	Status	Comments	Reviewed By	Approved By	Sign-off	Date
A. Map Frame Elements – Layer & Data Frame Properties						
All layers display correctly with no broken data sources	✓ Approved	All layers linked to bluesky.gdb	TZ	IDM	TZ, AS, IDM	04/12/2026
Layer names are clear, descriptive and free of underscores	✓ Approved	Display names cleaned for final layouts	TZ	IDM	TZ, AS, IDM	04/12/2026
Layer visibility toggled appropriately for intended purpose	✓ Approved	Only relevant layers turned on for each map	TZ	IDM	TZ, AS, IDM	04/12/2026
Scale dependent rendering configured where appropriate	✓ Approved	Applied where needed for community-scale maps	TZ	IDM	TZ, AS, IDM	04/12/2026
B. Layout Frame Elements – Cartographic Elements						
Layout page size appropriate (Letter 8.5×11 or specified size)	✓ Approved	Layouts exported at standard page size	IDM	AS	TZ, AS, IDM	04/12/2026
Layout orientation appropriate (Portrait or Landscape)	✓ Approved	Orientation matches map content	IDM	AS	TZ, AS, IDM	04/12/2026
North Arrow present, correctly oriented, and appropriately sized	✓ Approved	North arrow placed and scaled correctly	IDM	AS	TZ, AS, IDM	04/12/2026
Scale Bar present, correct units, and proportionate to map extent	✓ Approved	Scale bar matches 3TM map units	IDM	AS	TZ, AS, IDM	04/12/2026
Scale Text included (e.g., 1:5,000)	✓ Approved	Scale text included on final layouts	IDM	AS	TZ, AS, IDM	04/12/2026
Legend present, clearly readable and matches all visible layers	✓ Approved	Legend shows visible layers only	IDM	AS	TZ, AS, IDM	04/12/2026
Map Title is clear, informative, and communicates purpose	✓ Approved	Titles describe theme and study area	IDM	AS	TZ, AS, IDM	04/12/2026
All layout elements are properly aligned and not overlapping	✓ Approved	Elements aligned with no overlap	IDM	AS	TZ, AS, IDM	04/12/2026
Author name(s) included on layout	✓ Approved	Group authors listed on layouts	IDM	AS	TZ, AS, IDM	04/12/2026
Date of creation included on layout	✓ Approved	Creation date included	IDM	AS	TZ, AS, IDM	04/12/2026
Coordinate system / projection listed on layout	✓ Approved	Projection listed as NAD 1983 3TM 114W	IDM	AS	TZ, AS, IDM	04/12/2026
Map frame border / neatline present and professional	✓ Approved	Neatline added for clean layout	IDM	AS	TZ, AS, IDM	04/12/2026
Map exported to PDF at appropriate resolution (≥ 150 dpi)	✓ Approved	PDF exported at submission quality	IDM	AS	TZ, AS, IDM	04/12/2026
Map layout PDF opens without errors and renders all elements correctly	✓ Approved	PDF checked and opens correctly	IDM	AS	TZ, AS, IDM	04/12/2026
Data sources / credits listed on layout	✓ Approved		IDM	AS	TZ, AS, IDM	04/12/2026

C. General Document & Metadata Properties							
Readme file created for the project	✓ Approved	README saved with project package	AS	TZ	TZ, AS, IDM	04/12/2026	
Metadata created for the project	✓ Approved	Project metadata completed	AS	TZ	TZ, AS, IDM	04/12/2026	
Project folder structure is clean and organized	✓ Approved	Folders organized as commondata, p20, and p30	AS	TZ	TZ, AS, IDM	04/12/2026	
No temporary / scratch layers left in the project	✓ Approved	No scratch layers in final project views	AS	TZ	TZ, AS, IDM	04/12/2026	
D. Symbolology & Labelling							
Symbolology is appropriate and relevant to each feature type	✓ Approved	Symbolology matches data type and purpose	TZ	IDM	TZ, AS, IDM	04/12/2026	
Colour scheme is clear, accessible and not overcrowded	✓ Approved	Colour ramps clear and readable	TZ	IDM	TZ, AS, IDM	04/12/2026	
Labels applied where features benefit from identification	✓ Approved	Labels added where useful	TZ	IDM	TZ, AS, IDM	04/12/2026	
Label font size and style is legible at output scale	✓ Approved	Label size readable at export scale	TZ	IDM	TZ, AS, IDM	04/12/2026	
No duplicate or conflicting labels	✓ Approved	Duplicate labels reviewed and removed	TZ	IDM	TZ, AS, IDM	04/12/2026	
Underscores removed from all visible text (layer names, labels)	✓ Approved	Underscores removed from visible text	TZ	IDM	TZ, AS, IDM	04/12/2026	
TOTAL ITEMS APPROVED		29	0	--- Auto-counted from Status column			

GEOS410 Group 1 – Data Integrity & Manipulation Final Deliverables QA/QC Checklist							
BLUE SKY NAD 1983 3TM 114W							
Reviewers: Teng Zhang, Apram Singh & Ifechukwudelu Maduekwe Date: 12/04/2026							
Section / Item	Status	Comments	Reviewed By	Approved By	Sign-off	Date	
A. ArcGIS Pro Project Deliverables							
ArcGIS Pro project (.aprx) opens without errors	✓ Approved	Project_BlueSky.aprx tested before submission	TZ	IDM	TZ, AS, IDM	04/12/2026	
Project saved with a clear, descriptive name	✓ Approved	Project file name is clear and descriptive	TZ	IDM	TZ, AS, IDM	04/12/2026	
B. Assignment Deliverables							
Document title page with student names	✓ Approved	Title page includes group member names	AS	IDM	TZ, AS, IDM	04/12/2026	
QA/QC checklists in MS Excel format	✓ Approved	Checklist workbook saved in Excel format	AS	IDM	TZ, AS, IDM	04/12/2026	
PDF map layout	✓ Approved	Final map layouts exported as PDF	AS	IDM	TZ, AS, IDM	04/12/2026	
KML file	✓ Approved	KML deliverable included in submission	IDM	AS	TZ, AS, IDM	04/12/2026	
Comparison chart of ArcGIS vs FME conversion included	✓ Approved	Comparison chart included in report	IDM	AS	TZ, AS, IDM	04/12/2026	
Screenshots of both ArcGIS and FME conversion processes & brief summary included	✓ Approved	Screenshots and summary included in conversion report	IDM	AS	TZ, AS, IDM	04/12/2026	
Inconversion steps report	✓ Approved						
TOTAL ITEMS APPROVED		8	0	--- Auto-counted from Status column			

Appendix D: GEOS 418 Data Dictionary

GEOS 419 Group 1 | Blue Sky: Calgary Community Livability Index

This document describes tables, fields, data types, constraints, and definitions used in the Calgary Community Livability Index geodatabase. Tables are linked through Comm_name as the common foreign key, with Communities_Final serving as the central consolidated table.

Communities_Final

Description: Primary consolidated table containing community-level attributes, break-and-enter statistics, park coverage, housing affordability metrics, transit coverage, and the computed livability composite index for each Calgary community.

Primary Key: OBJECTID (PK) | Foreign Key: Comm_name (FK links to all related tables)

Field Name	Data Type	Constraints	Description
OBJECTID	INT	PK, NOT NULL	Unique system-generated identifier for each community feature.
Comm_name	VARCHAR	FK, NOT NULL	Name of the community; used as the foreign key linking to all related tables.
Comm_code	VARCHAR	NOT NULL	Standardized short code assigned to each community.
Sector	VARCHAR		City sector in which the community is located (e.g., North, South, Centre).
Srg	VARCHAR		Statistical research geography grouping for the community.
Comm_struc	VARCHAR		Community structure classification (e.g., established, developing).
Comm_pop_2025	FLOAT		Estimated community population for the year 2025.
Comm_area	FLOAT		Total geographic area of the community (square meters).
BE_Rate_Per_1000	INT		Break-and-enter incident rate per 1000 residents.

Field Name	Data Type	Constraints	Description
Ward	VARCHAR		Municipal ward designation for the community.
Police_District	VARCHAR		Calgary Police Service district is responsible for the community.
Total_BE	INT		Total number of break-and-enter incidents recorded in the community.
Commerical_BE	INT		Number of break-and-enter incidents at commercial premises.
Dwelling_BE	INT		Number of break-and-enter incidents at residential dwellings.
Parks_area	FLOAT		Total area of parkland within the community (square meters).
Park_perc	FLOAT		Percentage of community area designated as parkland.
Median_price_2025	FLOAT		Median residential property sale price in the community for 2025 (CAD).
Household_income_2025	FLOAT		Median household income in the community for 2025 (CAD).
House_price_to_income	FLOAT		Ratio of median house price to median household income; affordability indicator.
Trans_cov_prec	FLOAT		Percentage of community area covered by transit service areas (bus).
LRT_cov_perc	FLOAT		Percentage of community area covered by LRT (CTrain) service areas.
Coverage_area_m2	FLOAT		Total transit coverage area within the community (square meters).
Total_cov_perc	FLOAT		Combined percentage of community area covered by all transit service types.

Field Name	Data Type	Constraints	Description
Index_raw	FLOAT		Unscaled composite livability index value before normalization.
Index	FLOAT		Normalized composite livability index score (0–40 scale).
Index_rank	INT		Rank of the community based on the normalized index (40 = best).
Index_pctl	FLOAT		Percentile position of the community within the index distribution.
Shape_length	FLOAT		Perimeter length of the community polygon geometry (meters).
Shape_area	FLOAT		Area of the community polygon geometry (square meters).

YYC_BE_Total

Description: Stores aggregated break-and-enter crime counts for each Calgary community, categorized by premises type (commercial, dwelling, other). Linked to Communities_Final via community name.

Primary Key: OBJECTID (PK) | Foreign Key: Comm_name (FK → Communities_Final)

Field Name	Data Type	Constraints	Description
OBJECTID	INT	PK, NOT NULL	Unique system-generated identifier for each record.
Comm_name	VARCHAR	FK, NOT NULL	Community name; foreign key linking to Communities_Final.
Ward	VARCHAR		Municipal ward designation for the community.
Police_district	VARCHAR		Calgary Police Service district covering the community.
Total_BE	INT		Total number of break-and-enter incidents in the community.

Field Name	Data Type	Constraints	Description
Commerical_BE	INT		Number of break-and-enter incidents at commercial premises.
Dwelling_BE	INT		Number of break-and-enter incidents at residential dwellings.
Other_premises_BE	INT		Number of break-and-enter incidents at other premise types (e.g., sheds, garages).

Parks

Description: Contains spatial and attribute data for parkland associated with each community, including park area and percentage coverage. Used to calculate the parks component of the livability index.

Primary Key: OBJECTID (PK) | Foreign Key: Comm_name (FK → Communities_Final)

Field Name	Data Type	Constraints	Description
OBJECTID	INT	PK, NOT NULL	Unique system-generated identifier for each park polygon feature.
Comm_name	VARCHAR	FK, NOT NULL	Community name; foreign key linking to Communities_Final.
Comm_code	VARCHAR		Standardized short code for the community.
Sector	VARCHAR		City sector in which the community is located.
Srg	VARCHAR		Statistical research geography grouping.
Comm_struc	VARCHAR		Community structure classification.
Comm_area	FLOAT		Total area of the community (square meters).
Parks_area	FLOAT		Total area of parkland within the community (square meters).
Park_perc	FLOAT		Percentage of community area designated as parkland.
Shape_length	FLOAT		Perimeter length of the park polygon geometry (meters).

Field Name	Data Type	Constraints	Description
Shape_area	FLOAT		Area of the park polygon geometry (square meters).

Transit_servicearea

Description: Represents the service area polygons (walkable catchment zones) generated around bus and CTrain stops. Used to compute transit coverage percentages per community.

Primary Key: OBJECTID (PK) | Foreign Key: Comm_name (FK → Communities_Final)

Field Name	Data Type	Constraints	Description
OBJECTID	INT	PK, NOT NULL	Unique system-generated identifier for each transit service area polygon.
Comm_name	VARCHAR	FK, NOT NULL	Community name; foreign key linking to Communities_Final.
Comm_code	VARCHAR		Standardized short code for the community.
FacilityID	INT		Identifier of the transit stop or facility that generated the service area.
Shape	TEXT		Geometry type descriptor for the service area polygon.
FromBreak	FLOAT		Lower bound of the distance or time break used to define the service area (meters or minutes).
ToBreak	FLOAT		Upper bound of the distance or time break used to define the service area (meters or minutes).
Shape_length	FLOAT		Perimeter length of the service area polygon (meters).
Shape_area	FLOAT		Area of the service area polygon (square meters).

LRT_servicearea

Description: Represents the service area polygons (walkable catchment zones) generated specifically around CTrain (LRT) stations. Used to calculate the LRT coverage component of the livability index.

Primary Key: OBJECTID (PK) | Foreign Key: Comm_name (FK → Communities_Final)

Field Name	Data Type	Constraints	Description
OBJECTID	INT	PK, NOT NULL	Unique system-generated identifier for each LRT service area polygon.
Comm_name	VARCHAR	FK, NOT NULL	Community name; foreign key linking to Communities_Final.
Comm_code	VARCHAR		Standardized short code for the community.
FacilityID	INT		Identifier of the LRT station that generated the service area.
Shape	TEXT		Geometry type descriptor for the service area polygon.
FromBreak	FLOAT		Lower bound of the distance or time break used to define the LRT service area (meters or minutes).
ToBreak	FLOAT		Upper bound of the distance or time break used to define the LRT service area (meters or minutes).
Shape_length	FLOAT		Perimeter length of the LRT service area polygon (meters).
Shape_area	FLOAT		Area of the LRT service area polygon (square meters).

YYC_MedianPrice_2025

Description: Contains 2025 residential real estate pricing data and household income figures per community, supporting the housing affordability component of the livability index.

Primary Key: OBJECTID (PK) | Foreign Key: Comm_name (FK → Communities_Final)

Field Name	Data Type	Constraints	Description
OBJECTID	INT	PK, NOT NULL	Unique system-generated identifier for each record.
Comm_name	VARCHAR	FK, NOT NULL	Community name; foreign key linking to Communities_Final.
Comm_code	VARCHAR		Standardized short code for the community.
Median_price_2025	FLOAT		Median residential property sale price in the community for 2025 (CAD).
Household_income_2025	FLOAT		Median household income in the community for 2025 (CAD).
House_price_to_income	FLOAT		Ratio of median house price to median household income; housing affordability indicator.

Comm_compositeIndex

Description: Stores the intermediate preprocessed indicator values and final composite livability index scores used for ranking communities. Each indicator is preprocessed (normalized/rescaled) before being combined into the index.

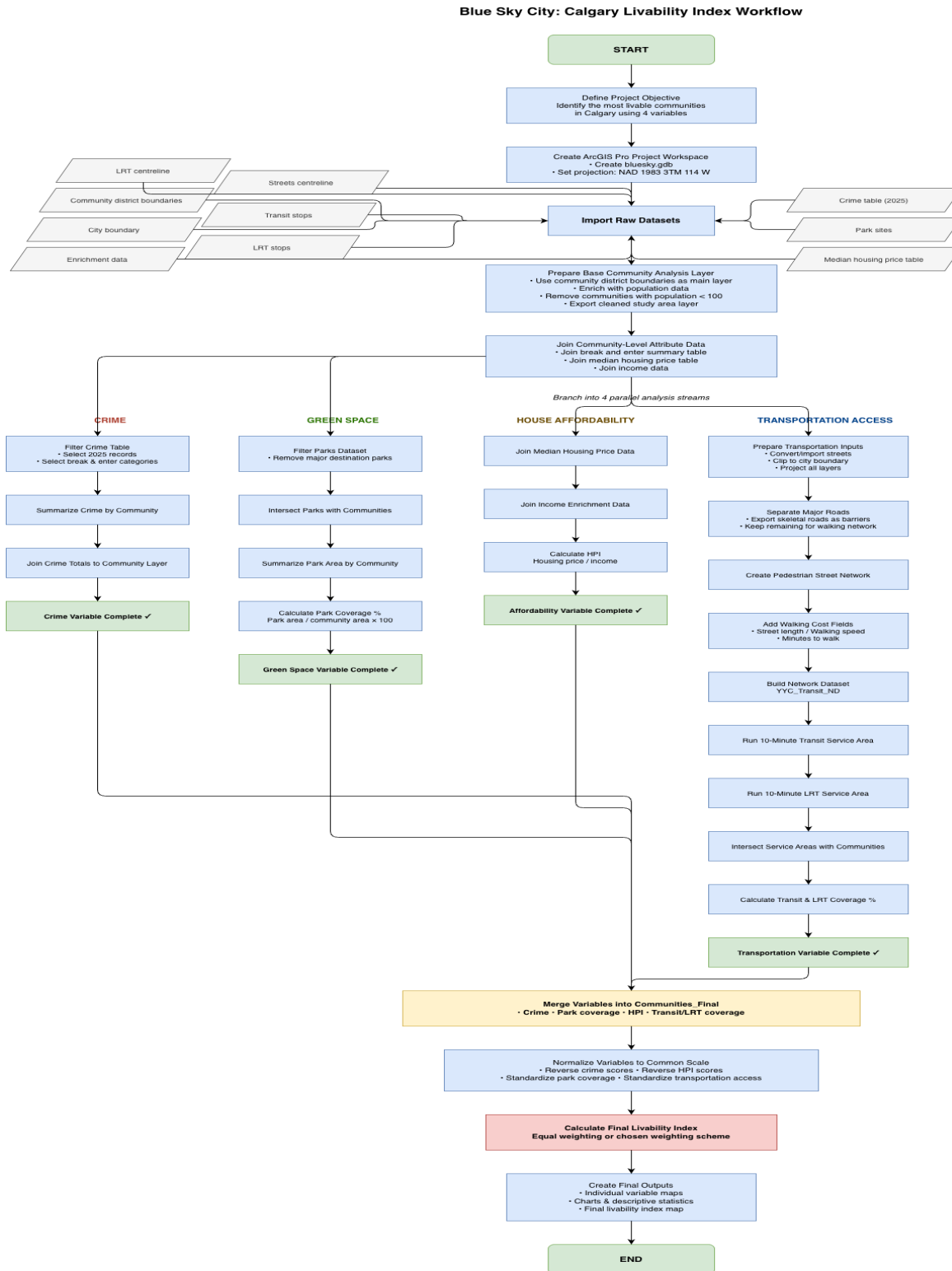
Primary Key: OBJECTID (PK) | Foreign Key: Comm_name (FK → Communities_Final)

Field Name	Data Type	Constraints	Description
OBJECTID	INT	PK, NOT NULL	Unique system-generated identifier for each record.
Comm_name	VARCHAR	FK, NOT NULL	Community name; foreign key linking to Communities_Final.
Shape	TEXT		Geometry type descriptor for the community polygon.
Source_id	INT		Reference ID linking back to the source dataset.

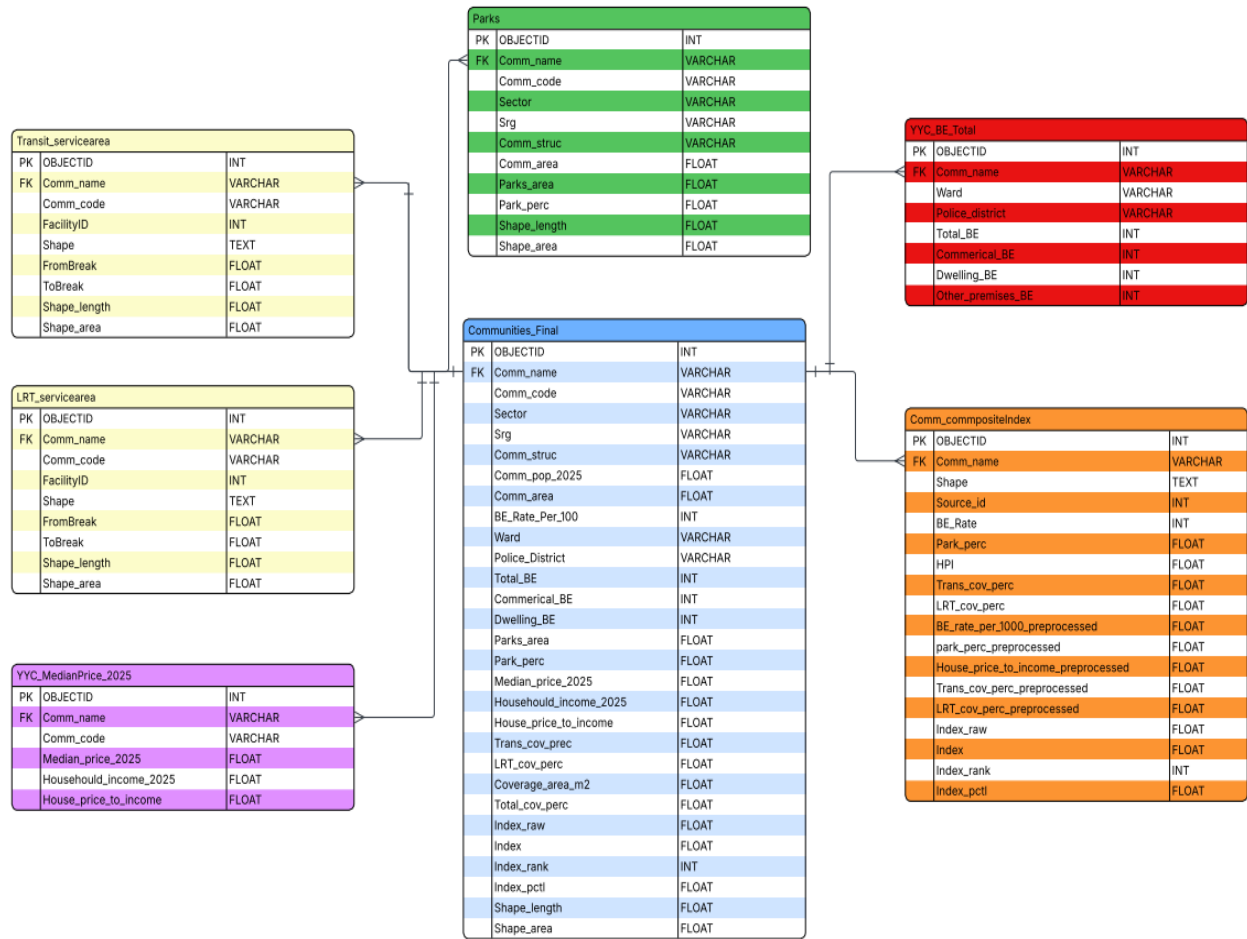
Field Name	Data Type	Constraints	Description
BE_Rate	INT		Raw break-and-enter rate per 1000 residents, prior to preprocessing.
Park_perc	FLOAT		Raw park percentage of community area, prior to preprocessing.
HPI	FLOAT		Raw house price-to-income ratio (Housing Price Index), prior to preprocessing.
Trans_cov_perc	FLOAT		Raw transit coverage percentage, prior to preprocessing.
LRT_cov_perc	FLOAT		Raw LRT coverage percentage, prior to preprocessing.
BE_rate_per_1000_preprocessed	FLOAT		Normalized break-and-enter rate (per 1,000 residents) after preprocessing for index calculation.
park_perc_preprocessed	FLOAT		Normalized park percentage value after preprocessing for index calculation.
House_price_to_income_preprocessed	FLOAT		Normalized house price-to-income ratio after preprocessing for index calculation.
Trans_cov_perc_preprocessed	FLOAT		Normalized transit coverage percentage after preprocessing for index calculation.

Field Name	Data Type	Constraints	Description
LRT_cov_perc_preprocessed	FLOAT		Normalized LRT coverage percentage after preprocessing for index calculation.
Index_raw	FLOAT		Unscaled composite livability index value computed from preprocessed indicators.
Index	FLOAT		Final normalized composite livability index score (0–40 scale).
Index_rank	INT		Community rank based on the normalized index (40 = highest livability).
Index_pctl	FLOAT		Percentile position of the community within the index score distribution.

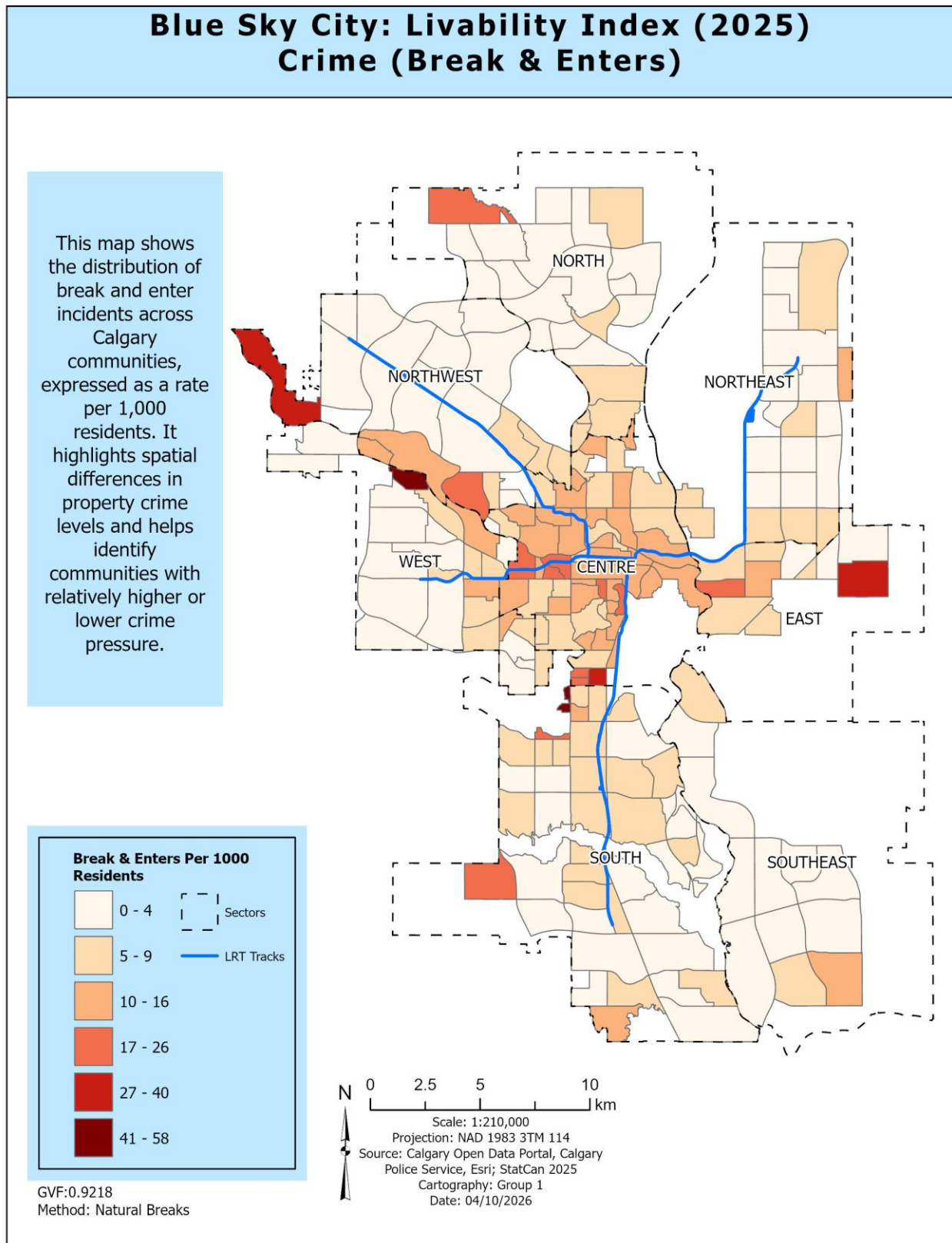
Appendix E: Project Methodology Flow Chart



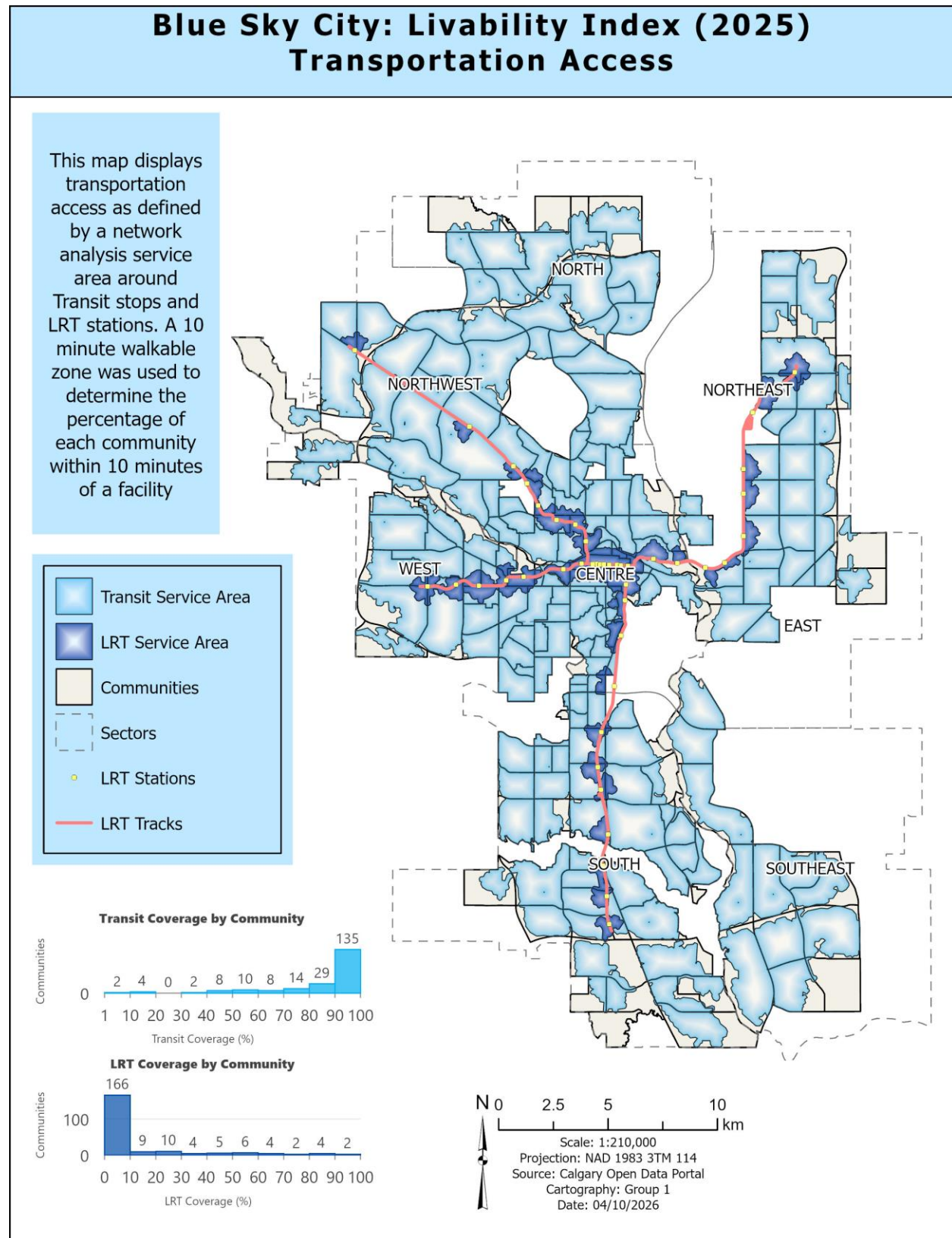
Appendix F: Entity Relationship Diagram



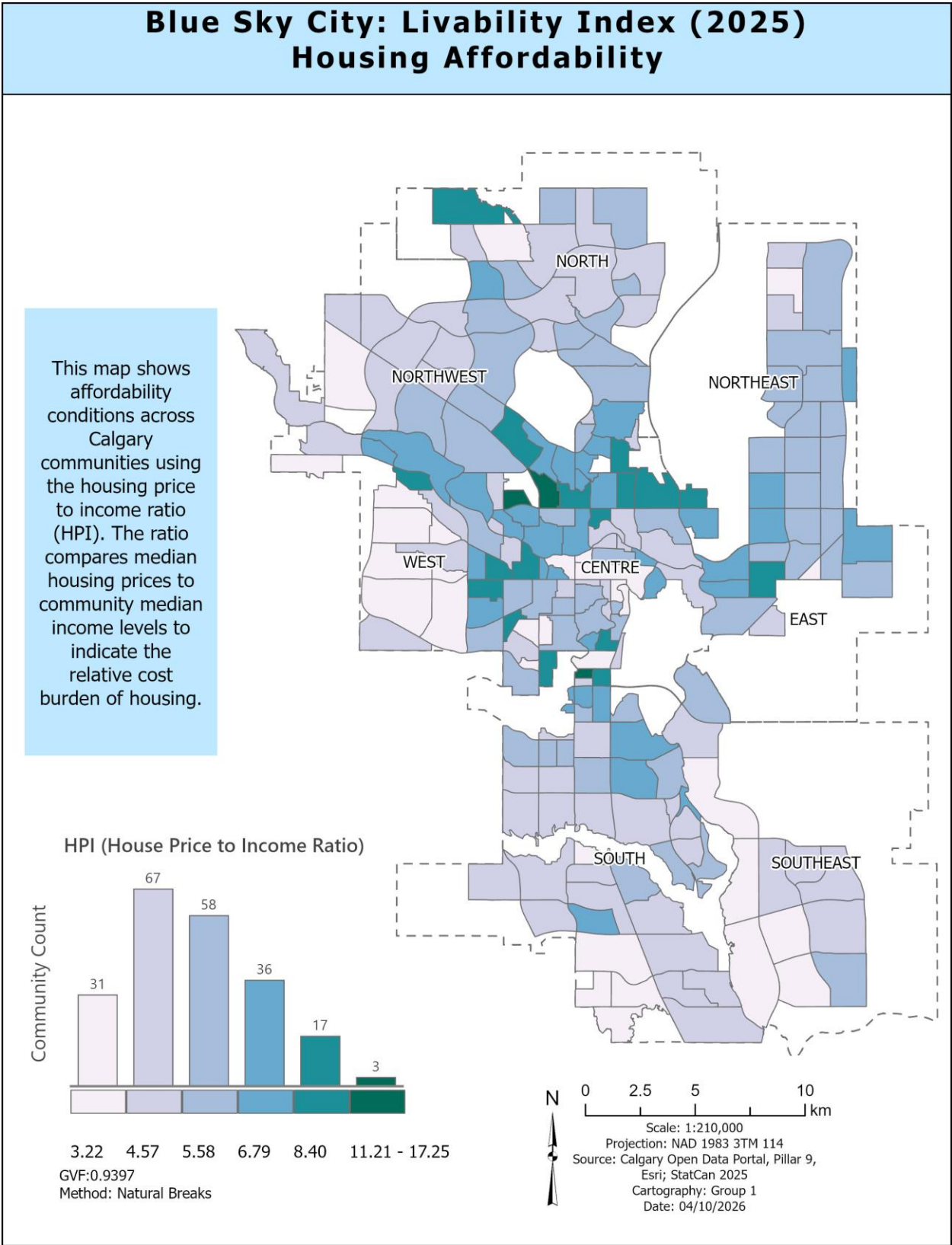
Appendix G: Crime (Break& Enter) Map



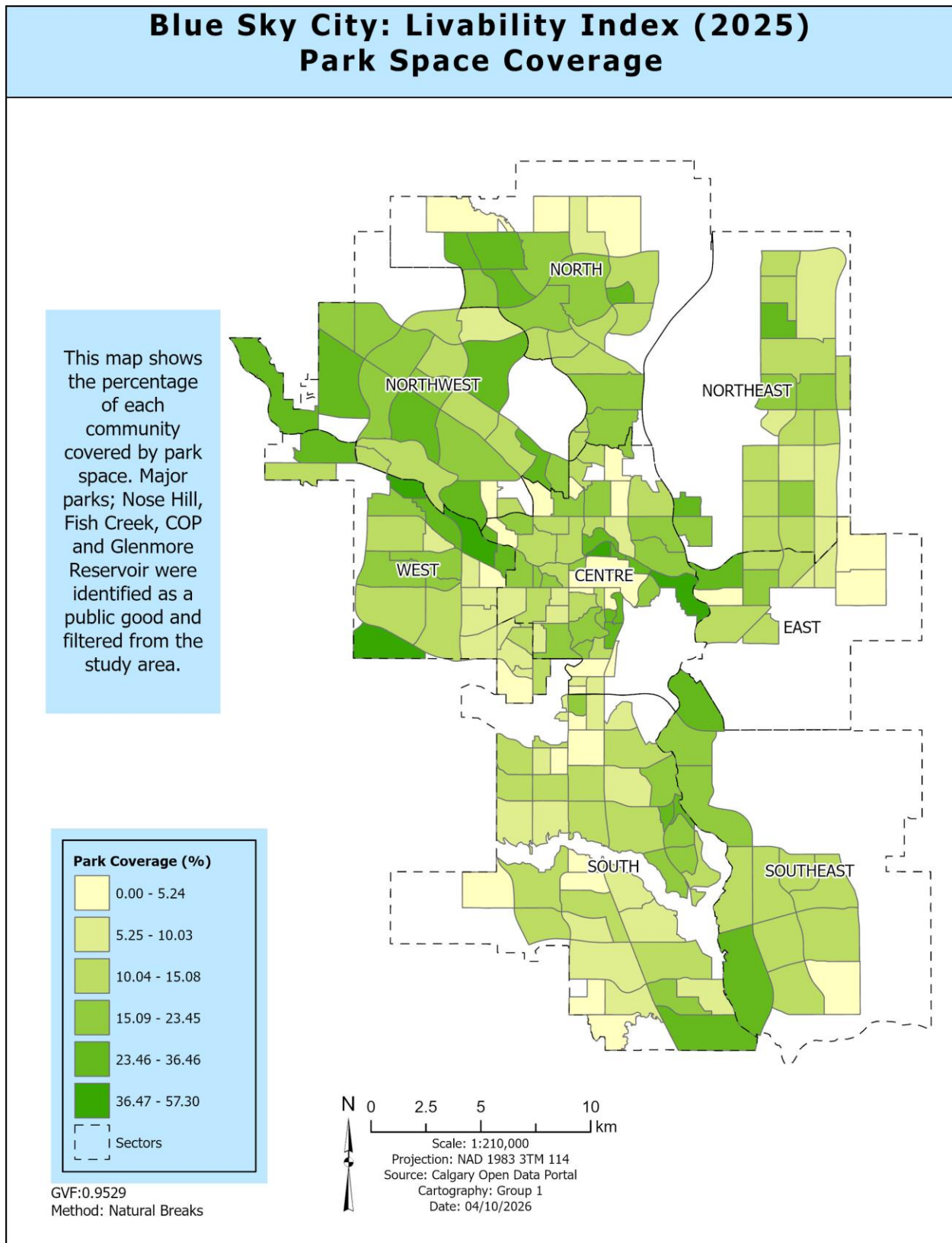
Appendix H: Transportation Access Map



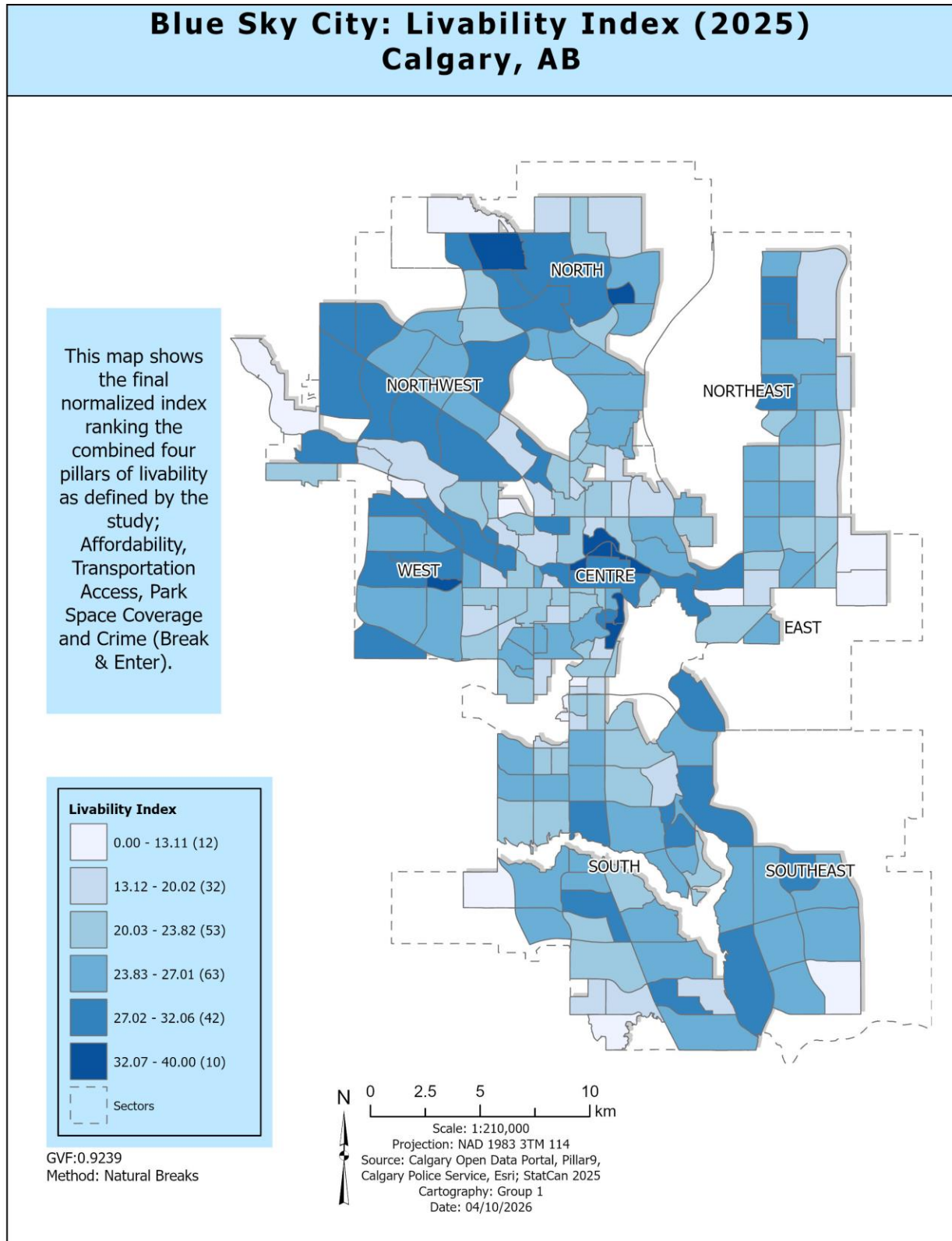
Appendix I: Housing Price to Income (HPI) Map



Appendix J: Park Space Coverage Map



Appendix K: Calgary Livability Index Map



Appendix L: Hotspot Analysis

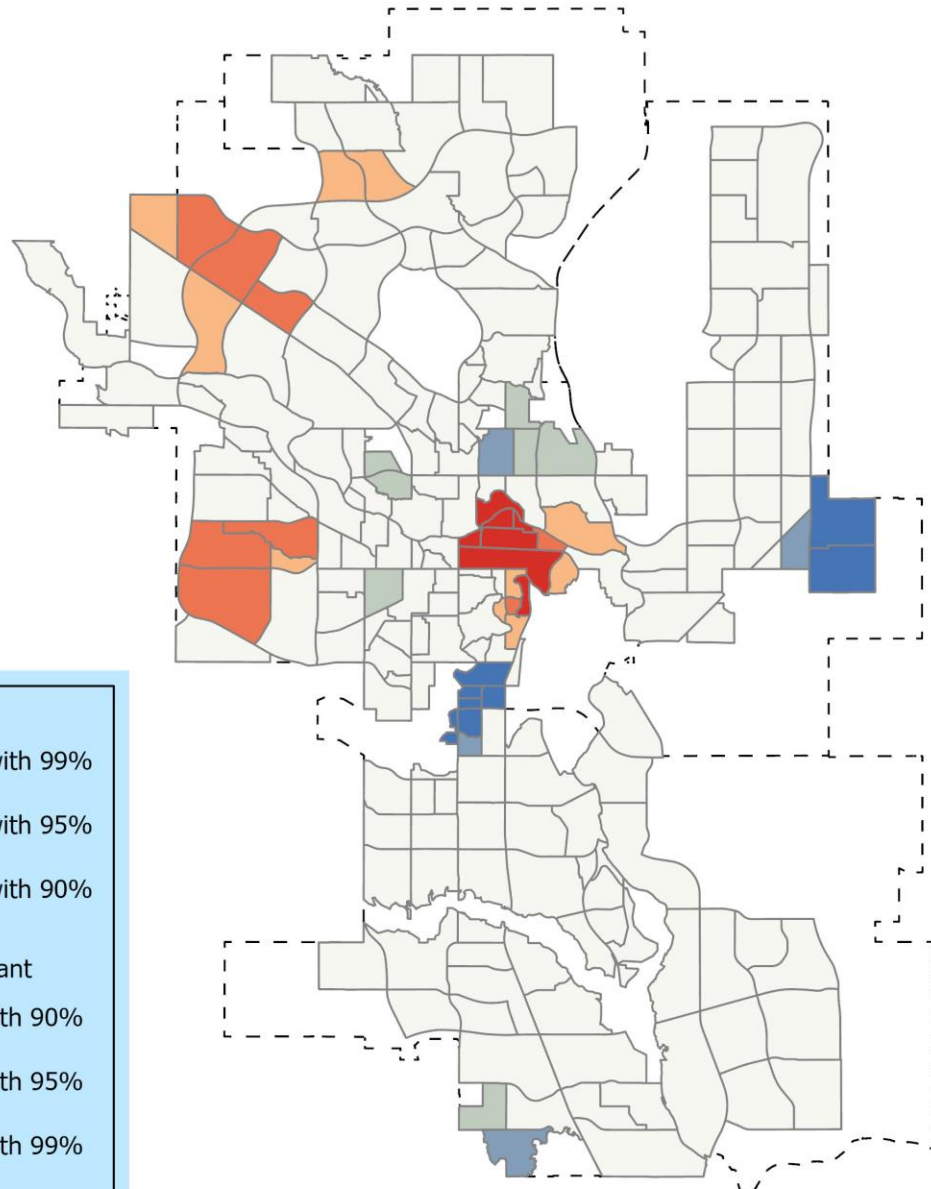
Blue Sky City: Livability Index (2025) Hotspot Analysis

This map identifies existing hotspots for the Calgary Livability Index. It shows clusters of significant high livability scores in red and significant clusters of low livability scores in blue, with varying confidence levels.

Getis-Ord Gi*

- Cold Spot with 99% Confidence
- Cold Spot with 95% Confidence
- Cold Spot with 90% Confidence
- Not Significant
- Hot Spot with 90% Confidence
- Hot Spot with 95% Confidence
- Hot Spot with 99% Confidence

- Sectors
- Community Boundary



N 0 2.5 5 10 km
 Scale: 1:210,000
 Projection: NAD 1983 3TM 114
 Source: Calgary Open Data Portal
 Cartography: Group 1
 Date: 04/10/2026

Appendix M: Cluster and Outlier Analysis Map

